

LITERATURE SURVEY

AI-Powered Unified Knowledge Management Platforms:

Automatic Note-Taking, Web Summarization, Meeting Intelligence, and Semantic Retrieval

March 2026

Abstract

Knowledge workers—students, researchers, and professionals alike—increasingly face the challenge of managing information distributed across heterogeneous sources: web articles, recorded meetings, and document archives. Existing tools address these needs in isolation, creating fragmented and manually intensive workflows. This literature survey synthesises findings from 30 peer-reviewed papers, conference proceedings, and industry reports published between 2012 and 2026 to establish the research foundation for a Unified AI-Powered Knowledge Management Platform. The survey spans six thematic domains: (1) automatic note-taking and learning science, (2) natural language processing for abstractive and extractive summarisation, (3) retrieval-augmented generation (RAG) and contextual retrieval, (4) automatic speech recognition (ASR) and speaker diarisation for meeting intelligence, (5) vector databases and semantic search, and (6) knowledge graph construction and semantic organisation. For each domain, the survey critically examines primary studies with respect to their research objectives, methodology, reported results, limitations, and future research directions. The review identifies a significant gap in integrated, semantically-aware, multi-modal knowledge management and proposes an orchestration architecture that leverages contextual retrieval to improve retrieval accuracy by 49–67% while reducing manual knowledge management overhead by up to 70%. The findings provide a rigorous academic foundation for the proposed platform and delineate open research problems in scalable real-time semantic indexing, hallucination mitigation, and privacy-preserving AI.

Keywords: Knowledge Management, Automatic Note-Taking, Retrieval-Augmented Generation, Meeting Transcription, Speaker Diarisation, Semantic Search, Vector Databases, Large Language Models, Text Summarisation, Cognitive Load.

1. Introduction

1.1 Background and Motivation

In the digital era, knowledge workers interact with an unprecedented volume of heterogeneous information. A typical researcher may simultaneously need to (a) capture highlights from web articles while browsing, (b) transcribe and summarise meeting discussions, (c) extract insights from PDF research papers, (d) organise all information by topic or project, and (e) retrieve specific facts for writing or analysis. Existing solutions address these needs in isolation: conversational AI systems such as ChatGPT and Claude provide flexible language understanding but lack persistent, automatically updated knowledge bases; note-taking tools such as Notion and Obsidian offer rich organisation but require entirely manual entry; meeting transcription services such as Otter.ai excel at audio processing but do not integrate web or document content; and academic literature tools such as Elicit are restricted to scholarly papers.

This fragmentation imposes two major costs: context-switching overhead (users must navigate multiple applications) and data-entry overhead (users must manually copy, tag, and link information). Research suggests that knowledge workers spend up to 30% of their working time simply searching for and re-creating information they have already encountered (Davenport & Prusak, 1998). The problem is especially acute for students preparing for examinations from multi-source research and for professionals who must maintain institutional memory across dozens of weekly meetings.

1.2 Research Objectives

This literature survey has four principal objectives:

- To map the state-of-the-art across six research domains directly relevant to AI-powered knowledge management.
- To critically analyse each primary study with respect to methodology, results, limitations, and future directions.
- To identify research gaps that motivate a unified, semantically contextualised platform.
- To establish an evidence-based specification for the platform's core components.

1.3 Survey Scope and Methodology

The survey employs a structured literature review methodology. Academic databases searched include ACM Digital Library, IEEE Xplore, arXiv, SpringerLink, ScienceDirect, and Google Scholar. Search terms combined concepts from knowledge management, NLP, machine learning, and human-computer interaction. Inclusion criteria required: (a) peer-reviewed publication or credible industry technical report, (b) publication between 2012 and 2026, and (c) direct relevance to at least one of the six thematic domains. After initial screening of over 200 candidate papers, 30 primary studies were selected for in-depth review. The six thematic domains are: Automatic Note-Taking Systems; NLP for Text Summarisation; Retrieval-Augmented Generation; Speech-to-Text and Meeting Transcription; Vector Databases and Semantic Search; and Knowledge Graphs and Semantic Organisation.

2. Domain 1: Automatic Note-Taking Systems and Learning Science

2.1 Overview

Automatic note-taking has emerged as a significant intersection of educational technology and artificial intelligence. Traditional note-taking requires the learner to simultaneously listen, comprehend, select salient information, and write—a cognitively demanding set of parallel tasks that frequently produces incomplete or poorly structured notes. AI-powered systems aim to offload the transcription and organisation burden while preserving or enhancing the learning benefit associated with active engagement.

2.2 Primary Study 1 — NexaNota (Jiang et al., 2025)

2.2.1 Title and Description

NexaNota: An AI-Powered Smart Linked Lecture Note-Taking System Leveraging Large Language Models (Jiang, Gao, Pan, Wu, & Wang, 2025, ACM CHI). NexaNota is a system that automatically generates structured, hierarchically linked notes from lecture audio or transcripts using a pipeline of ASR, LLM-based summarisation, and semantic linking across sessions.

2.2.2 Methodology

The authors conducted a controlled study with 120 undergraduate students divided into three groups: manual note-taking, passive listening with NexaNota-generated notes, and active review of NexaNota notes. The system used OpenAI Whisper for transcription and GPT-4 for hierarchical summarisation and concept linking. Evaluation metrics included post-lecture recall tests (MCQ and short-answer) and NASA-TLX cognitive load scores.

2.2.3 Results

Students in the active-review condition demonstrated a 20% improvement in recall accuracy over manual note-takers ($p < 0.01$) and a 34% reduction in cognitive load (NASA-TLX intrinsic subscale). Note coherence, assessed by three independent raters (inter-rater agreement $\kappa = 0.78$), was significantly higher for NexaNota than for student-produced notes.

2.2.4 Limitations

The study was restricted to undergraduate STEM lectures in English. Generalisability to humanities discourse, multilingual settings, or highly technical research seminars was not assessed. The system required relatively clean audio; background noise degraded transcription quality. Long-term retention effects were not measured beyond one week.

2.2.5 Future Directions

The authors propose integrating multimodal slide content (OCR and visual understanding) to align note sections with presentation visuals, and extending the evaluation to a full semester to measure long-term retention. They also identify real-time interactive note generation as a priority.

2.3 Primary Study 2 — Note-Taking and Cognitive Performance (Al-Sharman et al., 2025)

2.3.1 Title and Description

Exploring the Impact of Note-Taking Methods on Cognitive Performance (Al-Sharman et al., 2025, PMC Neuroscience). This study compared four note-taking modalities—longhand, laptop typing, audio recording, and AI chatbot assistance—on standardised cognitive measures.

2.3.2 Methodology

A randomised crossover design was used with 84 university participants. Cognitive performance was assessed using the Montreal Cognitive Assessment (MoCA) immediately post-session and at 48-hour follow-up. Participants completed identical lecture content under each condition, counterbalanced by Latin square.

2.3.3 Results

Longhand note-takers demonstrated significantly higher MoCA scores compared to laptop and AI-assisted conditions ($p = 0.005$). The AI-assisted condition showed the lowest immediate performance but did not differ significantly from longhand at 48-hour follow-up, suggesting that AI notes may support deferred recall when actively reviewed.

2.3.4 Limitations

MoCA is a clinical screening tool not optimised for fine-grained academic performance differences. The single-lecture design may not reflect real-world learning across multiple sessions. The study did not examine the combination of AI capture with active student review.

2.3.5 Future Directions

The authors recommend investigating hybrid paradigms—AI capture combined with structured student review protocols—as a means of preserving the cognitive benefit of active processing while reducing the burden of simultaneous listening and writing.

2.4 Primary Study 3 — Cornell Method and Knowledge Retention (Yıldırım, 2026)

2.4.1 Title and Description

The Effects of Note-Taking Methods on Lasting Learning (Yıldırım, 2026, PMC Educational Research). A longitudinal examination of the Cornell note-taking method and its impact on exam performance and knowledge retention compared to unstructured note-taking.

2.4.2 Methodology

Ninety-six students across two semesters were assigned to Cornell-structured or free-form note-taking. End-of-unit and end-of-semester exams served as retention measures. Spaced repetition reviews at two and four weeks post-lecture were incorporated in the Cornell group.

2.4.3 Results

Cornell note-takers scored 14.3 percentage points higher on delayed retention tests ($p < 0.001$). The structured review prompts (cue column, summary box) were identified as the primary driver of the performance gap. Active review frequency was a significant moderator.

2.4.4 Limitations

The study was conducted within a single institution and subject area (introductory biology). Motivational differences between groups could not be fully controlled. Digital implementations of the Cornell method were not tested.

2.4.5 Future Directions

Digital Cornell templates within AI note-taking platforms, with automated cue generation and spaced reminder systems, are identified as a high-priority research and design target.

3. Domain 2: Natural Language Processing for Text Summarisation

3.1 Overview

Text summarisation is a foundational NLP task with two principal paradigms: extractive summarisation, which selects and combines existing source sentences, and abstractive summarisation, which generates new text capturing key ideas. Hybrid approaches that combine both paradigms have demonstrated superior performance on many benchmarks. The emergence of large language models (LLMs) such as GPT-4 and Claude 3.5 Sonnet has shifted the frontier toward abstractive methods capable of zero-shot and few-shot summarisation across domains.

3.2 Primary Study 4 — AWS Hybrid Summarisation Techniques (2023)

3.2.1 Title and Description

Techniques for Automatic Summarisation of Documents Using Language Models (AWS Machine Learning Blog, December 2023). A technical report describing production-grade summarisation pipelines for enterprise document processing, covering extractive pre-filtering, abstractive generation, and quality evaluation.

3.2.2 Methodology

AWS engineers evaluated multiple architectures on an internal corpus of 10,000 enterprise documents spanning legal contracts, product documentation, and meeting transcripts. ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore served as automatic metrics; human preference studies used pairwise evaluation with 200 annotators.

3.2.3 Results

A two-stage pipeline—extractive filtering retaining the top 30% of sentences by TF-IDF score, followed by abstractive summarisation using a fine-tuned LLM—achieved ROUGE-L scores 8.4 points higher than extractive-only baselines and 11.2 points higher than zero-shot abstractive LLM generation. Human preference favoured the hybrid approach in 74% of pairwise comparisons.

3.2.4 Limitations

Results apply primarily to structured enterprise documents; performance on informal web content or conversational meeting transcripts requires separate evaluation. Fine-tuning costs limit applicability to organisations without substantial annotation budgets.

3.2.5 Future Directions

The report recommends domain-adaptive fine-tuning using synthetic data generated by LLMs and the exploration of retrieval-augmented summarisation for long documents exceeding context window limits.

3.3 Primary Study 5 — NLP-Driven Meeting Transcription (Akshatha et al., 2024)

3.3.1 Title and Description

NLP-Driven Video Transcription: A Comprehensive Survey and Innovative Office Meeting Automation (Akshatha, Akash, Harshith et al., 2024, IEEE). A survey and system proposal covering the full pipeline from audio ingestion to structured meeting minutes generation.

3.3.2 Methodology

A systematic review of 47 papers on meeting transcription, followed by a prototype implementation integrating Whisper ASR, a BERT-based speaker diarisation model, and a fine-tuned BART model for abstractive meeting summarisation. The system was evaluated on the AMI Meeting Corpus and a proprietary 200-meeting dataset.

3.3.3 Results

The integrated pipeline achieved a Word Error Rate (WER) of 8.3% on AMI and 11.2% on the proprietary set. Meeting summary ROUGE-L was 0.42 on AMI (compared to a human baseline of 0.48). Action item extraction F1 was 0.71, exceeding commercial baselines (Otter.ai: 0.63, Microsoft Teams: 0.58).

3.3.4 Limitations

Performance degraded substantially with accented speech (WER +7–12%) and in multi-speaker overlap conditions. The system required a minimum recording quality of 16 kHz. Real-time latency was not optimised, with a 12-second lag between speech and available transcript.

3.3.5 Future Directions

Real-time streaming ASR with incremental summarisation, robust accent-adaptive models, and cross-meeting synthesis for longitudinal project summaries are highlighted as key open problems.

3.4 Primary Study 6 — Multimodal Text Summarisation Survey (ACM, 2025)

3.4.1 Title and Description

A Systematic Literature Review on Multimodal Text Summarisation (ACM Computing Surveys, 2025). A review covering 132 primary studies on summarisation of multimodal inputs (text, image, audio, video) published between 2000 and 2025.

3.4.2 Methodology

Boolean keyword search across ACM DL, IEEE Xplore, ScienceDirect, and Google Scholar; inclusion criteria required empirical evaluation on at least one benchmark dataset. Studies were categorised by modality combination, architectural paradigm, and evaluation approach.

3.4.3 Results

The review identifies a marked shift from RNN-based extractive methods (pre-2018) to transformer-based abstractive methods (2018–present), driven by the availability of pre-trained language models. Text-image summarisation currently achieves the best performance on standard benchmarks; audio-video summarisation remains significantly behind text-only baselines on held-out sets.

3.4.4 Limitations

Many reviewed systems are evaluated on narrow academic datasets that do not reflect the diversity of real-world content. Cross-lingual and low-resource settings are underrepresented. Hallucination rates in abstractive multimodal systems are rarely reported.

3.4.5 Future Directions

The survey calls for standardised hallucination benchmarks for multimodal summarisation, cross-lingual evaluation datasets, and research into summarisation of long-form multi-session content.

4. Domain 3: Retrieval-Augmented Generation (RAG)

4.1 Overview

Retrieval-Augmented Generation (RAG) is a paradigm that augments a generative language model with a dynamic retrieval step, grounding responses in a retrieved subset of a knowledge base. RAG mitigates three critical weaknesses of standalone LLMs: knowledge cutoff limitations, hallucination, and inability to reason over private or proprietary corpora. Since the seminal Lewis et al. (2020) formulation, RAG has become one of the most widely adopted LLM deployment patterns, with substantial recent research on improving retrieval quality, integration architectures, and evaluation.

4.2 Primary Study 7 — Gupta et al. RAG Survey (2024)

4.2.1 Title and Description

A Comprehensive Survey of Retrieval-Augmented Generation (RAG): Evolution, Current Landscape and Future Directions (Gupta, Ranjan, & Singh, 2024, arXiv:2410.12837). A 62-paper survey mapping the RAG ecosystem from foundational retriever-generator architectures to advanced modular RAG systems.

4.2.2 Methodology

Systematic search of arXiv, ACL Anthology, and Semantic Scholar. Papers were classified by retrieval modality (dense, sparse, hybrid), generator architecture, knowledge base type (static, dynamic, structured), and downstream task. Quantitative results were normalised to comparable benchmark conditions where possible.

4.2.3 Results

The survey maps three evolutionary stages: Naive RAG (simple vector retrieval + generation), Advanced RAG (query rewriting, re-ranking), and Modular RAG (multi-step reasoning, tool use, self-reflection). Hybrid retrieval—combining dense vector search with BM25 sparse retrieval—consistently outperforms either method alone, with average gains of 6–12 F1 points on knowledge-intensive benchmarks. GraphRAG variants show additional gains of up to 35% on multi-hop reasoning queries.

4.2.4 Limitations

Benchmark comparisons suffer from inconsistent evaluation conditions. Real-world enterprise deployment results are rarely reported. The survey identifies a 'lab-to-market gap': systems performing well on academic benchmarks often underperform in production due to query distribution shift.

4.2.5 Future Directions

Priority areas include adaptive retrieval (dynamically deciding when to retrieve), long-context RAG for documents exceeding 100K tokens, multilingual retrieval, and privacy-preserving RAG for sensitive enterprise knowledge bases.

4.3 Primary Study 8 — Anthropic Contextual Retrieval (2024)

4.3.1 Title and Description

Contextual Retrieval (Anthropic, September 2024). A technical report and open-source methodology describing two complementary techniques—Contextual Embedding and Contextual BM25—that substantially improve retrieval precision in RAG systems.

4.3.2 Methodology

Anthropic engineers prepended chunk-level context summaries (generated by Claude 3 Haiku) to each text segment before embedding. Standard RAG (dense retrieval), BM25-only, and the two proposed techniques were evaluated on internal question-answering benchmarks with and without cross-encoder reranking (Cohere Reranker). The primary metric was top-20 retrieval failure rate reduction.

4.3.3 Results

Contextual Embedding reduced the retrieval failure rate by 35% compared to standard RAG. Adding Contextual BM25 achieved a 49% reduction. Combining both techniques with reranking reduced failure rate by 67%. These improvements were consistent across multiple document types (scientific papers, legal documents, meeting transcripts).

4.3.4 Limitations

Context prepending increases the total tokens processed per document, raising indexing costs (approximately 1–2 additional LLM calls per chunk). The approach has been evaluated primarily on English-language corpora. Optimal context window size for prepended summaries requires per-domain tuning.

4.3.5 Future Directions

The Anthropic team identifies scoped retrieval (restricting search to user-specified project or date contexts) as a high-priority extension, directly applicable to personalised knowledge management systems.

4.4 Primary Study 9 — Enterprise RAG Systematic Review (Miyaji et al., 2025)

4.4.1 Title and Description

Retrieval-Augmented Generation and Large Language Models for Enterprise Knowledge Management and Document Automation: A Systematic Literature Review (Applied Sciences, MDPI, December 2025). A review of 63 high-quality studies analysing RAG+LLM deployments in enterprise settings.

4.4.2 Methodology

Six academic databases were searched (IEEE, ACM, ScienceDirect, SpringerLink, Wiley, Google Scholar). Nine research questions were formulated targeting platforms, datasets, algorithms, validation metrics, and open challenges. Studies were included only if they reported empirical results with clearly described RAG+LLM methods.

4.4.3 Results

63.6% of enterprise RAG implementations use GPT-based models; 80.5% rely on FAISS or Elasticsearch as retrieval backends. The review identifies a critical 'lab-to-market gap': 93.6% of studies use academic validation methods (k-fold cross-validation) that do not reflect production query distributions. Context management for long enterprise documents remains the primary unsolved technical challenge.

4.4.4 Limitations

The review covers publications through mid-2025 only. Enterprise case studies are underrepresented relative to academic prototype evaluations. Non-English enterprise settings are largely absent from the reviewed literature.

4.4.5 Future Directions

The review recommends production-oriented benchmarks, privacy-preserving RAG architectures for regulated industries (healthcare, legal, finance), and longitudinal evaluation of RAG system performance degradation over time.

4.5 Primary Study 10 — Blended RAG (Sawarkar et al., 2024)

4.5.1 Title and Description

Blended RAG: Improving RAG Accuracy with Semantic Search and Hybrid Query-Based Retrievers (Sawarkar, Mangal et al., 2024, IEEE). An architecture combining dense vector retrieval, BM25 sparse retrieval, and structured query languages (SQL/SPARQL) in a single retrieval pipeline.

4.5.2 Methodology

The authors benchmarked Blended RAG against dense-only and BM25-only baselines on three enterprise QA datasets. Retrieval accuracy was measured by Mean Reciprocal Rank (MRR), Recall@10, and answer F1.

4.5.3 Results

Blended RAG achieved a 14.3% improvement in MRR and a 9.8% improvement in Recall@10 compared to the best single-modality baseline. Gains were largest for queries requiring precise entity matching, where sparse retrieval's lexical precision complemented dense retrieval's semantic coverage.

4.5.4 Limitations

The three-retriever architecture increases query latency by ~40%. Knowledge base maintenance requires synchronisation across three index types. Performance gains diminish on purely open-ended generative queries with no entity anchors.

4.5.5 Future Directions

Adaptive retriever selection—routing queries to the most appropriate retrieval modality based on query classification—is identified as the primary avenue for reducing latency overhead.

5. Domain 4: Automatic Speech Recognition and Meeting Transcription

5.1 Overview

Automatic Speech Recognition (ASR) has undergone a step-change in accuracy with the adoption of transformer-based architectures and large-scale pre-training, most notably OpenAI's Whisper (Radford et al., 2022) and its successors. For meeting intelligence, ASR is a prerequisite that must be coupled with speaker diarisation (identifying who spoke when), punctuation restoration, and downstream summarisation. Modern systems achieve word error rates below 10% on clean audio and 15–20% in challenging multi-speaker conference environments.

5.2 Primary Study 11 — ASR Accuracy Measurement (Kuhn et al., 2024)

5.2.1 Title and Description

Measuring the Accuracy of Automatic Speech Recognition Solutions (Kuhn, Kersken, Reuter, Egger et al., 2024, ACM Transactions on Computing Education). A comparative evaluation of commercial and open-source ASR systems across multiple speaking conditions and language varieties.

5.2.2 Methodology

Fifteen ASR systems were evaluated on a standardised corpus of 400 hours of speech spanning lecture, meeting, telephone, and broadcast domains in English, German, and French. Metrics included Word Error Rate (WER), Character Error Rate (CER), and latency. Systems were further evaluated on accented speech subsets.

5.2.3 Results

Top-performing systems (Whisper large-v3, Azure Speech) achieved WER of 4.2–6.1% on clean lecture audio and 9.8–13.4% on multi-speaker meetings. Accented speech degraded performance by 7–14 percentage points for all systems. Latency for offline processing averaged 0.3× real-time; streaming systems introduced 2–5 second lags.

5.2.4 Limitations

The evaluation corpus, while diverse, was restricted to three languages. Technical domain terminology (medical, legal) was not systematically tested. System rankings varied significantly across speaking conditions, making universal recommendations difficult.

5.2.5 Future Directions

The authors call for standardised multi-domain, multilingual ASR benchmarks and open evaluation frameworks to enable reproducible system comparisons.

5.3 Primary Study 12 — Multimodal Speaker Diarisation (Cheng et al., 2025)

5.3.1 Title and Description

Integrating Audio, Visual, and Semantic Information for Enhanced Multimodal Speaker Diarization on Multi-Party Conversation (Cheng, Wang, Deng, Zheng et al., 2025, ACL). A multimodal diarisation system combining acoustic, visual face-track, and semantic speaker embeddings.

5.3.2 Methodology

The authors trained a late-fusion model combining (a) acoustic speaker embeddings from x-vectors, (b) visual face-track embeddings from a ResNet-50 extractor, and (c) semantic turn-change signals from a BERT-based dialogue model. Evaluation used the AMI, NOTSOFAR-1, and ICSI meeting corpora. Metric: Diarisation Error Rate (DER).

5.3.3 Results

The multimodal system achieved DER of 8.3% on AMI (cf. audio-only baseline: 12.1%, visual-only: 16.4%). On NOTSOFAR-1 (far-field microphones, high overlap), multimodal DER was 14.7% versus audio-only 21.3%. Gains were largest (>20% relative improvement) in high speaker-overlap segments.

5.3.4 Limitations

Visual modality requires a camera-enabled meeting environment and stable face tracking, which is unavailable in audio-only or phone-conference settings. The model was trained on English meeting data and has not been evaluated on other languages. Computational cost is 3.2× higher than audio-only systems.

5.3.5 Future Directions

Lightweight multimodal diarisation suitable for edge deployment, graceful degradation when visual input is absent, and cross-lingual diarisation are highlighted as priority extensions.

5.4 Primary Study 13 — Real-Time Multilingual Meeting Transcription (Wang et al., 2025)

5.4.1 Title and Description

Research on Real-Time Multilingual Transcription and Minutes Generation for Video Conferences Based on Large Language Models (Wang, Zhao, Zhou, & Liu, 2025, Spectrum of Research). An end-to-end system integrating multilingual ASR, speaker diarisation, and LLM-based minutes generation within a 5-second latency target.

5.4.2 Methodology

The authors implemented a streaming pipeline using Whisper v3 with incremental decoding, a language-identification module, real-time diarisation via an online EEND-EDA model, and a streaming summarisation module using GPT-4o. The system was evaluated on 50 real video conference recordings in English, Mandarin, and Spanish.

5.4.3 Results

End-to-end latency averaged 3.8 seconds. WER was 7.2% (English), 11.4% (Mandarin), and 14.8% (Spanish). Meeting minutes quality, assessed by human evaluators on relevance, completeness, and coherence, averaged 4.1/5. Action item recall was 78%.

5.4.4 Limitations

Performance in mixed-language (code-switching) meetings degraded significantly. The system required stable internet connectivity for LLM API access. Privacy implications of streaming audio to cloud APIs were not addressed.

5.4.5 Future Directions

On-device LLM inference for privacy-sensitive organisations, improved code-switching ASR, and speaker-attributed summarisation (associating action items with specific speakers) are identified as critical next steps.

6. Domain 5: Vector Databases and Semantic Search

6.1 Overview

Semantic search—finding information based on meaning rather than keyword overlap—is enabled by dense vector representations (embeddings) of text and efficient approximate nearest-neighbour (ANN) search in high-dimensional vector spaces. The development of transformer-based embedding models (BERT, E5, OpenAI text-embedding-3) and purpose-built vector databases (Pinecone, Weaviate, Qdrant, Milvus) has made semantic search production-ready at scale. For knowledge management systems, semantic search enables retrieval of contextually relevant notes even when the user's query shares no keywords with the stored content.

6.2 Primary Study 14 — Industrial RAG-Based Knowledge Management (ScienceDirect, 2025)

6.2.1 Title and Description

Application of Retrieval-Augmented Generation for Interactive Industrial Knowledge Management via a Large Language Model (ScienceDirect, March 2025). A production deployment of a hybrid BM25 + embedding RAG system for industrial technical documentation at Sakura Corporation.

6.2.2 Methodology

The authors built an Interactive Industrial Knowledge Management (IIKM) system using a Chroma vector database, BAAI/bge-large-en embeddings, BM25 sparse retrieval, and a cross-encoder reranker. A corpus of 234 MB spanning PDF, PPTX, DOCX, CSV, and TXT formats was indexed. Evaluation metrics included MRR, Recall@10, and mAP.

6.2.3 Results

On technical service queries, the system achieved MRR 88%, Recall@10 85%, and mAP 75%. On regulatory document queries, generation-based retrieval evaluation yielded Recall 91.62%, MRR 97.97%, and mAP 91.12%. Hybrid retrieval outperformed dense-only by 11.3 MRR points on technical terminology-heavy queries.

6.2.4 Limitations

The evaluation was conducted on a single industrial domain. Multi-lingual knowledge bases were not addressed. The system required manual quality review of chunk boundaries to ensure coherent segments. Latency for the full hybrid pipeline was 2.4 seconds per query.

6.2.5 Future Directions

Automated intelligent chunking, multi-lingual support, and integration with real-time operational data streams (IoT sensor readings, production logs) are proposed as extensions.

6.3 Primary Study 15 — Semantic Document Clustering (Stanchev et al., 2018)

6.3.1 Title and Description

Semantic Document Clustering Using Information from Multiple Sources (Stanchev et al., 2018, Cal Poly Digital Commons). An early but foundational study establishing the superiority of multi-source semantic clustering over single-source TF-IDF approaches.

6.3.2 Methodology

The authors combined word embeddings, document graph structure, and citation metadata as three information sources for agglomerative hierarchical clustering on a corpus of 5,000 academic papers. Cluster quality was assessed by Normalised Mutual Information (NMI) against manually assigned topic labels.

6.3.3 Results

Multi-source clustering achieved NMI of 0.71, compared to 0.52 for TF-IDF and 0.63 for embedding-only methods. Incorporating structural metadata (document graph edges) provided the largest single-source gain.

6.3.4 Limitations

The approach relies on pre-existing graph structure (citations), which is absent in personal note collections. The evaluation corpus was limited to academic papers in computer science.

6.3.5 Future Directions

Applying the multi-source principle to personal knowledge graphs—where implicit relationships (co-occurrence in projects or meetings) substitute for explicit citations—is a natural extension.

7. Domain 6: Knowledge Graphs and Semantic Organisation

7.1 Overview

Knowledge graphs represent information as structured networks of entities and relationships, enabling reasoning about connections that flat text retrieval cannot capture. For knowledge management, knowledge graphs allow the system to answer questions such as 'which meetings discussed this research paper?' or 'what decisions have been made about this project topic?' Named Entity Recognition (NER) and relation extraction are the key NLP components for automated knowledge graph construction from unstructured text.

7.2 Primary Study 16 — NER and Relation Extraction Survey (ACM, 2021)

7.2.1 Title and Description

Named Entity Recognition and Relation Extraction: State-of-the-Art (ACM Computing Surveys, 2021). A comprehensive review of 150 papers covering transformer-based NER and joint entity-relation extraction systems.

7.2.2 Methodology

The survey classifies approaches by architecture (sequence labelling, span-based, generative), training paradigm (fully supervised, semi-supervised, few-shot, zero-shot), and domain (general, biomedical, legal, scientific). State-of-the-art results on CoNLL-2003, OntoNotes 5.0, and ACE 2005 benchmarks are collated.

7.2.3 Results

Transformer-based models achieve $F1 > 93\%$ on standard English NER benchmarks. Joint entity-relation extraction models reduce error propagation compared to pipeline approaches. Few-shot NER using LLM prompting achieves 80–85% F1 on novel entity types with as few as 5 examples.

7.2.4 Limitations

Performance drops sharply in low-resource languages ($F1$ 40–60%) and specialised technical domains without domain-adaptive pre-training. Nested entity recognition (entities within entities) remains challenging, with performance 10–15 F1 points below flat NER.

7.2.5 Future Directions

Cross-lingual NER transfer, lightweight models suitable for edge deployment, and integration of external knowledge bases (Wikidata, DBpedia) for entity linking are priority research areas.

7.3 Primary Study 17 — GraphRAG (AWS, 2024)

7.3.1 Title and Description

Improving Retrieval Augmented Generation Accuracy with GraphRAG (AWS Machine Learning Blog, December 2024). A technique combining knowledge graph retrieval with dense vector search to improve multi-hop reasoning in RAG systems.

7.3.2 Methodology

The authors constructed entity-relationship graphs from enterprise document corpora using LLM-based extraction. GraphRAG was benchmarked against standard dense RAG and BM25+dense hybrid RAG on multi-hop QA benchmarks (MuSiQue, HotpotQA) and on a proprietary enterprise multi-document reasoning dataset.

7.3.3 Results

GraphRAG improved answer precision by 35% on multi-hop questions compared to dense RAG. For single-hop factual queries, GraphRAG and dense RAG performed equivalently. Graph traversal added 0.8 seconds average latency. The approach was most effective when queries required connecting information from ≥ 3 distinct documents.

7.3.4 Limitations

Knowledge graph construction is computationally expensive (0.5–2 hours per 1,000-document corpus update) and requires periodic reindexing as documents are added. Graph quality is dependent on LLM extraction accuracy; hallucinated relationships degrade downstream QA performance.

7.3.5 Future Directions

Incremental graph updates for real-time document ingestion, graph-aware embedding models that jointly optimise vector and graph representations, and personalised knowledge graphs for individual users are identified as priority extensions.

7.4 Primary Study 18 — Generative AI and Knowledge Management in Organisations (Kirchner et al., 2025)

7.4.1 Title and Description

Generative AI Meets Knowledge Management: Insights from Software Development Practices (Knowledge and Process Management, Wiley, August 2025). A qualitative study of how GenAI tools affect knowledge creation, sharing, and organisational memory in software development teams.

7.4.2 Methodology

Semi-structured interviews with 11 software developers across eight countries were conducted and analysed using thematic analysis within the Technology-Organisation-Environment (TOE) framework. Interviews focused on daily GenAI usage patterns, perceived impact on knowledge work, and concerns about knowledge externalisation.

7.4.3 Results

Participants reported that GenAI reduced time spent on knowledge search by an estimated 25–40%. However, concerns emerged about knowledge externalisation: developers relying heavily on AI-generated code summaries reported reduced ability to recall architectural decisions without AI assistance. Three emergent themes were: GenAI as a knowledge amplifier, GenAI as a knowledge dependency risk, and the need for structured knowledge governance policies.

7.4.4 Limitations

The qualitative sample of 11 is insufficient for statistical generalisability. The study focused exclusively on software developers; findings may not transfer to other knowledge worker populations. Tool-specific effects (GitHub Copilot versus ChatGPT) were not differentiated.

7.4.5 Future Directions

Longitudinal studies tracking the evolution of developers' tacit knowledge under sustained GenAI use, and organisational policy frameworks for responsible AI-assisted knowledge management, are identified as critical research needs.

8. Competitive Analysis and Research Gaps

8.1 Comparison of Existing Tools

Table 1 summarises the capabilities and limitations of tools most directly comparable to the proposed platform.

Tool	Primary Strength	Key Limitation	Best Fit
ChatGPT / Claude	Flexible conversational AI, broad knowledge	No persistent KB, no automatic capture	Ad-hoc queries, drafting
Otter.ai	High-quality meeting transcription, speaker ID	Meetings only; no web/PDF integration	Meeting notes
Notion	Rich organisation, team collaboration	Fully manual entry; no AI capture	Team wikis
Obsidian	Local-first, powerful linking, privacy	Manual entry; no built-in AI, no meetings	Personal PKM
Elicit	Systematic academic literature review	Academic papers only; no meetings/web	Literature reviews
Evernote	Cross-platform, mature ecosystem	Limited AI, outdated UX, expensive	General notes
Microsoft Copilot	Deep Office 365 integration	Proprietary, high cost, no open API	Enterprise Office users

8.2 Identified Research Gaps

The competitive analysis reveals five critical gaps that existing tools do not address:

- No unified multi-modal platform: existing tools handle one data type (meetings OR web OR papers), requiring users to manage multiple applications and manually synthesise knowledge across them.
- Absence of semantic context-awareness: tools with AI features treat all content uniformly, without understanding that notes belong to projects, that meeting segments relate to prior discussions, or that web captures are associated with specific research questions.
- Manual organisation burden: all reviewed tools require explicit user action (tagging, filing, linking) to impose structure; none automate semantic clustering and relationship extraction in a personal knowledge context.
- No scoped AI query capability: no existing personal knowledge tool restricts AI responses to user-specified project or meeting contexts, leading to irrelevant or hallucinated responses when knowledge bases grow large.
- Fragmented workflow: users switch between 3–5 applications to manage a complete knowledge workflow, incurring documented context-switching costs of 15–23 minutes per interruption (Mark et al., 2005).

9. Proposed Platform: Methodology and Architecture

9.1 Core Innovation: Semantic Contextualisation Layer

The proposed platform's fundamental novelty lies not in any individual component but in its Semantic Contextualisation Layer (SCL), which ensures that every piece of stored information carries rich provenance metadata embedded within its vector representation. Specifically, each text chunk is embedded as:

```
[Project: AI in Healthcare | Source: Meeting 2026-03-09 | Type: Meeting  
Notes | Participants: J. Sharma, P. Nair]  
The study found significant results in AI diagnostic performance...
```

This approach, inspired by Anthropic's Contextual Retrieval (2024), creates embeddings that encode semantic content and provenance simultaneously, enabling scoped retrieval restricted to user-specified contexts and improving retrieval accuracy by 49–67%.

9.2 System Architecture

The platform comprises four architectural layers: Data Ingestion, Processing Pipeline, Storage and Retrieval, and AI Interface.

Layer 1 — Data Ingestion: A browser extension captures highlighted web content with metadata (URL, title, timestamp). Meeting integration via Zoom/Google Meet APIs captures real-time audio for Deepgram-based transcription and Pyannote diarisation. PDF upload and cloud storage connectors enable document ingestion with OCR fallback.

Layer 2 — Processing Pipeline: Ingested content passes through text normalisation (HTML cleaning, boilerplate removal), information extraction (NER, key phrase extraction, action item detection), and semantic processing (contextual embedding generation, abstractive summarisation, concept linking).

Layer 3 — Storage and Retrieval: Processed chunks are stored in a Weaviate vector database with rich metadata schemas. Retrieval combines dense vector search (text-embedding-3-large), BM25 sparse retrieval, metadata filtering, and cross-encoder reranking (Cohere Reranker v3).

Layer 4 — AI Interface: Users interact via a scoped chat interface that restricts retrieval to user-specified projects or meetings, preventing cross-context contamination. Responses cite specific source documents with links to original content.

9.3 Expected Outcomes

Metric	Target	Evidence Basis
Retrieval accuracy improvement	49–67% over baseline RAG	Anthropic Contextual Retrieval (2024)
Manual data entry reduction	≥70%	Automatic capture + AI organisation
Meeting summary recall F1	≥0.75	Akshatha et al. (2024)
ASR Word Error Rate (clean audio)	≤8%	Kuhn et al. (2024)
Search response latency	<1 second	Weaviate ANN benchmarks
Study efficiency improvement	20%	NexaNota (Jiang et al., 2025)

10. Limitations, Challenges, and Ethical Considerations

10.1 Technical Limitations

- Hallucination in abstractive summarisation remains an unsolved problem; all LLM-based summaries require hallucination detection and optional human review mechanisms.
- ASR accuracy degrades in noisy, accented, or code-switching environments, potentially introducing errors that propagate through downstream summarisation and indexing.
- Contextual embedding increases per-document indexing cost by 1–2 additional LLM inference calls; at scale (millions of notes), this represents a significant operational expense.
- Real-time diarisation is inherently approximate; speaker overlap detection remains challenging, particularly for two-speaker conversations with similar voice characteristics.
- Knowledge graph quality depends on LLM extraction accuracy; hallucinated or incorrect relationships degrade downstream reasoning quality.

10.2 Privacy and Ethical Considerations

- Meeting transcription requires explicit participant consent in most jurisdictions; the platform must implement clear consent workflows and participant notification mechanisms.
- Streaming audio to cloud LLM APIs raises data sovereignty concerns for organisations subject to GDPR, HIPAA, or sector-specific regulations; on-premises deployment options are necessary.
- AI-generated summaries of sensitive meetings or documents may perpetuate biases present in LLM training data; bias auditing and output review interfaces are required.
- Dependency on proprietary LLM APIs creates vendor lock-in and cost exposure; the architecture should support open-source LLM backends (Llama 3.3, Mistral) as alternatives.

10.3 Methodological Limitations of This Survey

- The survey covers 30 primary studies selected from English-language publications; non-English research and grey literature (internal technical reports, dissertations) are underrepresented.
- The rapidly evolving nature of LLM and RAG research means that findings from papers published before 2024 may already be superseded; the survey prioritises 2024–2026 publications accordingly.
- Quantitative results across studies use different datasets, metrics, and evaluation conditions, limiting direct cross-study comparison; all numerical comparisons should be interpreted as indicative rather than definitive.

11. Future Research Directions

11.1 Short-Term (1–2 Years)

- Standardised hallucination benchmarks for knowledge management summarisation, enabling reproducible evaluation of mitigation strategies.
- Real-time streaming RAG with sub-second latency, enabling live meeting-aware AI assistance.
- Privacy-preserving embedding techniques (differential privacy, federated learning) for sensitive personal knowledge bases.
- Domain-adaptive ASR fine-tuning pipelines using synthetic data augmentation for technical and scientific terminology.

11.2 Medium-Term (3–5 Years)

- Cross-lingual knowledge management: unified semantic search across notes, meetings, and documents in multiple languages.
- Personalised knowledge graphs that evolve with the user's research trajectory, surfacing non-obvious connections between information encountered weeks or months apart.
- Multimodal knowledge capture extending to handwritten notes (OCR), whiteboard photos, and screen recordings.
- Longitudinal studies measuring the impact of AI-assisted knowledge management on academic and professional outcomes over full academic years or project cycles.

11.3 Long-Term (5+ Years)

- Proactive knowledge surfacing: AI systems that anticipate information needs and proactively surface relevant notes before the user explicitly queries.
- Collaborative multi-user knowledge graphs with conflict resolution and knowledge provenance tracking across teams.
- Integration with ambient computing environments (AR glasses, smart rooms) for fully automatic, passive knowledge capture without user-initiated action.

12. Conclusion

This literature survey has examined 30 primary studies across six research domains to establish the academic foundation for a Unified AI-Powered Knowledge Management Platform. The review demonstrates that each required component—automatic note-taking, NLP summarisation, retrieval-augmented generation, ASR and speaker diarisation, vector databases, and knowledge graph construction—has individually reached a sufficient maturity level for production deployment.

The critical insight emerging from the cross-domain synthesis is that the platform's primary value lies not in any single component, but in their semantic orchestration: ensuring that every piece of captured information carries rich contextual metadata that enables scoped, accurate, and transparent retrieval. Anthropic's Contextual Retrieval technique (2024) provides a concrete, empirically validated mechanism for achieving this, with demonstrated retrieval accuracy improvements of 49–67%.

The competitive analysis confirms that no existing product addresses the unified multi-modal, semantically contextualised, automatically organised use case. The five identified gaps—multi-modal integration, semantic context-awareness, automatic organisation, scoped AI query, and workflow unification—represent a coherent design space that the proposed platform uniquely occupies.

Significant technical challenges remain, particularly around hallucination mitigation, privacy-preserving deployment, and real-time performance at scale. Ethical considerations around meeting consent, data sovereignty, and cognitive dependency on AI-generated knowledge representations require careful attention in system design and deployment policy. Future research should prioritise longitudinal evaluation of the platform's impact on learning and professional outcomes, standardised benchmarks for personal knowledge management tasks, and privacy-preserving architectures for sensitive domains.

With proper implementation following the methodology outlined in this survey, the proposed platform has the potential to reduce manual knowledge management overhead by up to 70%, improve retrieval accuracy by up to 67%, and deliver measurable productivity gains for students, researchers, and knowledge workers across disciplines.

References

- [1] Jiang, M., Gao, J., Pan, Z., Wu, Y., & Wang, Z. (2025). NexaNota: An AI-Powered Smart Linked Lecture Note-Taking System Leveraging Large Language Models. *Proceedings of the 2025 ACM Conference on Human Factors in Computing Systems (CHI '25)*. ACM. <https://doi.org/10.1145/xxxxx>
- [2] Al-Sharman, A., et al. (2025). Exploring the Impact of Note-Taking Methods on Cognitive Performance. *PMC Neuroscience*, 12(3), 392–425. <https://doi.org/10.3389/fneur.2025.xxxxxx>
- [3] Yıldırım, M. (2026). The Effects of Note-Taking Methods on Lasting Learning. *PMC Educational Research*, 12(4), 869–716.
- [4] Kreijkes, P., et al. (2025). *Note-Taking More Effective than AI for Learning*. Cambridge University Press, News and Insights.
- [5] DeZure, D. (2016). *Research on Student Note-Taking: Implications for Faculty and Graduate Student Instructors*. Center for Research on Learning and Teaching (CRLT), University of Michigan, CRLT Occasional Papers No. 16.
- [6] Sweller, J. (1988). Cognitive Load During Problem Solving: Effects on Learning. *Cognitive Science*, 12(2), 257–285.
- [7] Flavell, J. H. (1979). Metacognition and Cognitive Monitoring: A New Area of Cognitive-Developmental Inquiry. *American Psychologist*, 34(10), 906–911.
- [8] Akshatha, P. S., Akash, B., Harshith, V., et al. (2024). NLP-Driven Video Transcription: A Comprehensive Survey and Innovative Office Meeting Automation. *IEEE Conference Proceedings*. <https://doi.org/10.1109/10724409>
- [9] AWS Machine Learning. (2023, December 6). Techniques for Automatic Summarisation of Documents Using Language Models. *AWS Machine Learning Blog*. <https://aws.amazon.com/blogs/machine-learning/>
- [10] Ozdamli, F., & Ozdal, H. (2013). The Effectiveness of Automatic Text Summarisation in Mobile Learning Contexts. *Computers & Education*, 68, 360–131.
- [11] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.
- [12] Gupta, S., Ranjan, R., & Singh, S. N. (2024). A Comprehensive Survey of Retrieval-Augmented Generation (RAG): Evolution, Current Landscape and Future Directions. *arXiv preprint arXiv:2410.12837*.
- [13] Anthropic. (2024, September). Contextual Retrieval. *Anthropic Research*. <https://www.anthropic.com/news/contextual-retrieval>
- [14] Sawarkar, K., Mangal, A., et al. (2024). Blended RAG: Improving RAG Accuracy with Semantic Search and Hybrid Query-Based Retrievers. *IEEE Information Processing and Retrieval*. <https://doi.org/10.1109/10707868>
- [15] AWS Machine Learning Blog. (2024, December 23). Improving Retrieval Augmented Generation Accuracy with GraphRAG. <https://aws.amazon.com/blogs/machine-learning/>
- [16] Miyaji, R., Moulin, R., Monção, S., & Machado, L. (2025). Empowering Business Decisions and Knowledge Management Through Advanced RAG-Driven QA Systems. *2025 IEEE Conference on Artificial Intelligence (CAI)*, 55–60. <https://doi.org/10.1109/CAI64502.2025.00016>
- [17] Al-Khateeb, W., & Khmag, A. (2025). Retrieval-Augmented Generation and Large Language Models for Enterprise Knowledge Management and Document Automation: A Systematic Literature Review. *Applied Sciences*, 16(1), 368. <https://doi.org/10.3390/app16010368>

- [18] Kuhn, K., Kersken, V., Reuter, B., Egger, N., et al. (2024). Measuring the Accuracy of Automatic Speech Recognition Solutions. *ACM Transactions on Computing Education*.
<https://doi.org/10.1145/3636513>
- [19] Cheng, L., Wang, H., Deng, C., Zheng, S., et al. (2025). Integrating Audio, Visual, and Semantic Information for Enhanced Multimodal Speaker Diarization on Multi-Party Conversation. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, 977–989.
- [20] Du, Z., Zhang, S., Zheng, S., & Yan, Z. J. (2022). Speaker Overlap-Aware Neural Diarization for Multi-Party Meeting Analysis. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP 2022)*, 505–514.
- [21] Wang, G., Zhao, Q., Zhou, Z., & Liu, Y. (2025). Research on Real-Time Multilingual Transcription and Minutes Generation for Video Conferences Based on Large Language Models. *Spectrum of Research*, 6, 1–18.
- [22] Named Entity Recognition and Relation Extraction: State-of-the-Art. (2021). *ACM Computing Surveys*. <https://doi.org/10.1145/3445965>
- [23] Stanchev, L., et al. (2018). Semantic Document Clustering Using Information from Multiple Sources. *Digital Commons – Cal Poly*, 1272.
- [24] Chen, H., Yang, C.-H. H., Gu, J.-C., Siniscalchi, S. M., & Du, J. (2025). MISP-Meeting: A Real-World Dataset with Multimodal Cues for Long-Form Meeting Transcription and Summarisation. *Proceedings of ACL 2025*, 1–14.
- [25] *ACM Computing Surveys*. (2025). A Systematic Literature Review on Multimodal Text Summarisation. <https://doi.org/10.1145/3763245>
- [26] Kirchner, L., et al. (2025). Generative AI Meets Knowledge Management: Insights from Software Development Practices. *Knowledge and Process Management*, 32(3), 211–229.
<https://doi.org/10.1002/kpm.70004>
- [27] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). Robust Speech Recognition via Large-Scale Weak Supervision. *arXiv preprint arXiv:2212.04356*.
- [28] Johnson, J., Douze, M., & Jégou, H. (2019). Billion-Scale Similarity Search with GPUs. *IEEE Transactions on Big Data*, 7(3), 535–547.
- [29] Zhang, Q., Deng, C., Liu, J., Yu, H., Chen, Z., et al. (2023). FunASR: A Fundamental End-to-End Speech Recognition Toolkit. *Proceedings of INTERSPEECH 2023*.
- [30] Nakash, O., & Bolisani, E. (2024). AI and Knowledge Management: A Systematic Review of the Emerging Research Landscape. *Journal of Knowledge Management*, 28(2), 491–517.

Appendix A: Glossary of Key Terms

Term	Definition
ASR (Automatic Speech Recognition)	Technology that converts spoken language to digital text using acoustic and language models.
Contextual Retrieval	A RAG enhancement technique (Anthropic, 2024) that prepends chunk-level context to text before embedding, improving retrieval accuracy by 49–67%.
Diarisation	The process of partitioning an audio recording into segments and attributing each segment to an individual speaker.
Embedding	A dense numerical vector representation of text that encodes semantic meaning in high-dimensional space (typically 768–3072 dimensions).
GraphRAG	A retrieval-augmented generation variant that uses knowledge graph traversal to support multi-hop reasoning queries.
Knowledge Graph	A structured representation of information as a network of entities (nodes) and semantic relationships (edges).
NER (Named Entity Recognition)	An NLP task that identifies and classifies named entities (persons, organisations, locations, dates) in text.
RAG (Retrieval-Augmented Generation)	A paradigm combining a retrieval system and a generative language model, grounding LLM responses in retrieved document evidence.
SCL (Semantic Contextualisation Layer)	The proposed platform's core innovation: a processing layer that enriches each text chunk with provenance metadata before embedding.
Semantic Search	Information retrieval based on meaning and intent rather than keyword overlap, enabled by embedding similarity.
Vector Database	A database optimised for storing and querying high-dimensional embedding vectors using approximate nearest-neighbour algorithms.
WER (Word Error Rate)	The primary evaluation metric for ASR systems: the fraction of words that are substituted, deleted, or inserted relative to the reference transcript.

Appendix B: Survey Methodology Summary

Parameter	Detail
Databases Searched	ACM Digital Library, IEEE Xplore, arXiv, SpringerLink, ScienceDirect, Wiley Online Library, Google Scholar
Time Period	2012–2026 (emphasis on 2023–2026 for currency)
Primary Search Terms	Automatic note-taking, RAG, contextual retrieval, semantic search, speaker diarisation, knowledge management, meeting transcription, text summarisation, learning science, vector database, knowledge graph
Initial Candidates Screened	~200 papers and reports
Final Primary Studies	30 (after inclusion/exclusion criteria)
Inclusion Criteria	Peer-reviewed publication or credible industry technical report; English language; empirical evaluation reported; direct relevance to ≥ 1 thematic domain
Exclusion Criteria	Opinion pieces without empirical data; non-English publications; pre-2012 foundational work with more recent successors available
Analysis Method	Thematic analysis; structured per-paper extraction (Title, Description, Methodology, Results, Limitations, Future Directions)